

1 **toExponential()**

- Converts a number into exponential notation.
- **Example:**
- `let num = 1234;`
- `console.log(num.toExponential(2)); // "1.23e+3"`

2 **isNaN()**

- Checks if a value is NaN (Not-a-Number).
- **Example:**
- `console.log(isNaN("123")); // false`
- `console.log(isNaN("abc")); // true`

3 **isFinite()**

- Checks if a value is a finite number.
- **Example:**
- `console.log(isFinite(123)); // true`
- `console.log(isFinite(Infinity)); // false`

4 **Math.round()**

- Rounds a number to the nearest integer.
- **Example:**
- `console.log(Math.round(4.5)); // 5`
- `console.log(Math.round(4.4)); // 4`

5 **Math.floor()**

- Rounds a number down to the nearest integer.
- **Example:**
- `console.log(Math.floor(4.7)); // 4`

6 **Math.ceil()**

- Rounds a number up to the nearest integer.
- **Example:**
- `console.log(Math.ceil(4.2)); // 5`

7 **Math.abs()**

- Returns the absolute value of a number.
- **Example:**
- `console.log(Math.abs(-5)); // 5`

8 **Math.pow()**

- Returns the base raised to the power of the exponent.
- **Example:**
- `console.log(Math.pow(2, 3)); // 8`

9 **Math.sqrt()**

- Returns the square root of a number.
- **Example:**
- `console.log(Math.sqrt(16)); // 4`

10 **Math.max()**

- Returns the largest of zero or more numbers.
- **Example:**
- `console.log(Math.max(1, 5, 3)); // 5`

11 **Math.min()**

- Returns the smallest of zero or more numbers.
- **Example:**
- `console.log(Math.min(1, 5, 3)); // 1`

12 **Math.random()**

- Returns a random number between 0 (inclusive) and 1 (exclusive).
- **Example:**
- `console.log(Math.random());` // Random number

13 **Math.trunc()**

- Removes the fractional part of a number.
- **Example:**
- `console.log(Math.trunc(4.9));` // 4

14 **Number.isInteger()**

- Checks if a value is an integer.
- **Example:**
- `console.log(Number.isInteger(5));` // true
- `console.log(Number.isInteger(5.5));` // false

15 **Number.isSafeInteger()**

- Checks if a number is a safe integer within JavaScript's limits.
- **Example:**
- `console.log(Number.isSafeInteger(123));` // true

JavaScript String Functions

21. **charAt()**

- Returns the character at a specified index.
- **Example:**
- `let str = "Hello";`
- `console.log(str.charAt(1));` // "e"

22. **charCodeAt()**

- Returns the Unicode value of the character at a specified index.
- **Example:**
- `console.log("A".charCodeAt(0));` // 65

23. **concat()**

- Joins two or more strings.
- **Example:**
- `let str1 = "Hello";`
- `let str2 = "World";`
- `console.log(str1.concat(" ", str2));` // "Hello World"

24. **includes()**

- Checks if a string contains a specified substring.
- **Example:**
- `console.log("Hello".includes("ell"));` // true

25. **indexOf()**

- Returns the index of the first occurrence of a specified value.
- **Example:**
- `console.log("Hello".indexOf("l"));` // 2

26. **lastIndexOf()**

- Returns the index of the last occurrence of a specified value.
- **Example:**
- `console.log("Hello".lastIndexOf("l"));` // 3

27. **toUpperCase()**

- Converts a string to uppercase.

- **Example:**
- `console.log("hello".toUpperCase()); // "HELLO"`

28.toLowerCase()

- Converts a string to lowercase.
- **Example:**
- `console.log("HELLO".toLowerCase()); // "hello"`

29.slice()

- Extracts a section of a string.
- **Example:**
- `let str = "Hello World";`
- `console.log(str.slice(0, 5)); // "Hello"`

30.substring()

- Extracts characters from a string between two indices.
- **Example:**
- `console.log("Hello".substring(1, 4)); // "ell"`

31.substr()

- Extracts characters from a string, starting at a specified index.
- **Example:**
- `console.log("Hello".substr(1, 3)); // "ell"`

32.replace()

- Replaces a specified value with another value in a string.
- **Example:**
- `console.log("Hello World".replace("World", "JavaScript")); // "Hello JavaScript"`

33.replaceAll()

- Replaces all occurrences of a substring.
- **Example:**
- `console.log("abcabc".replaceAll("abc", "123")); // "123123"`

34.split()

- Splits a string into an array of substrings.
- **Example:**
- `console.log("a,b,c".split(",")); // ["a", "b", "c"]`

35.trim()

- Removes whitespace from both ends of a string.
- **Example:**
- `console.log(" Hello ".trim()); // "Hello"`

36.padStart()

- Pads a string from the start to a specified length.
- **Example:**
- `console.log("5".padStart(3, "0")); // "005"`

37.padEnd()

- Pads a string from the end to a specified length.
- **Example:**
- `console.log("5".padEnd(3, "0")); // "500"`

38.startsWith()

- Checks if a string starts with a specified substring.
- **Example:**
- `console.log("Hello".startsWith("He")); // true`

39.endsWith()

- Checks if a string ends with a specified substring.

- **Example:**
- `console.log("Hello".endsWith("lo")); // true`

40. repeat()

- Repeats a string a specified number of times.
- **Example:**
- `console.log("Hi".repeat(3)); // "HiHiHi"`

JavaScript Array Functions

41. push()

- Adds elements to the end of an array.
- **Example:**
- `let arr = [1, 2];`
- `arr.push(3);`
- `console.log(arr); // [1, 2, 3]`

42. pop()

- Removes the last element from an array.
- **Example:**
- `let arr = [1, 2, 3];`
- `arr.pop();`
- `console.log(arr); // [1, 2]`

43. shift()

- Removes the first element from an array.
- **Example:**
- `let arr = [1, 2, 3];`
- `arr.shift();`
- `console.log(arr); // [2, 3]`

44. unshift()

- Adds elements to the beginning of an array.
- **Example:**
- `let arr = [2, 3];`
- `arr.unshift(1);`
- `console.log(arr); // [1, 2, 3]`

45. splice()

- Adds or removes elements at a specific index.
- **Example:**
- `let arr = [1, 2, 3];`
- `arr.splice(1, 1, "a", "b");`
- `console.log(arr); // [1, "a", "b", 3]`

46. slice()

- Returns a shallow copy of a portion of an array.
- **Example:**
- `let arr = [1, 2, 3, 4];`
- `console.log(arr.slice(1, 3)); // [2, 3]`

47. concat()

- Merges two or more arrays.
- **Example:**
- `let arr1 = [1, 2];`
- `let arr2 = [3, 4];`
- `console.log(arr1.concat(arr2)); // [1, 2, 3, 4]`

48. indexOf()

- Returns the first index of a value in an array.
- **Example:**
- `console.log([1, 2, 3].indexOf(2)); // 1`

49. lastIndexOf()

- Returns the last index of a value in an array.
- **Example:**
- `console.log([1, 2, 3, 2].lastIndexOf(2)); // 3`

50. includes()

- Checks if an array contains a specified value.
- **Example:**
- `console.log([1, 2, 3].includes(2)); // true`

51. join()

- Converts all elements of an array into a string, separated by a specified delimiter.
- **Example:**
- `let arr = [1, 2, 3];`
- `console.log(arr.join("-")); // "1-2-3"`

52. reverse()

- Reverses the order of elements in an array.
- **Example:**
- `let arr = [1, 2, 3];`
- `arr.reverse();`
- `console.log(arr); // [3, 2, 1]`

53. sort()

- Sorts the elements of an array in place.
- **Example:**
- `let arr = [3, 1, 2];`
- `arr.sort();`
- `console.log(arr); // [1, 2, 3]`

54. filter()

- Creates a new array with elements that pass the test implemented by the provided function.
- **Example:**
- `let arr = [1, 2, 3, 4];`
- `let even = arr.filter(num => num % 2 === 0);`
- `console.log(even); // [2, 4]`

55. map()

- Creates a new array populated with the results of calling a function on every element in the array.
- **Example:**
- `let arr = [1, 2, 3];`
- `let squared = arr.map(num => num * num);`
- `console.log(squared); // [1, 4, 9]`

56. reduce()

- Applies a function to reduce an array to a single value.
- **Example:**
- `let arr = [1, 2, 3, 4];`
- `let sum = arr.reduce((acc, val) => acc + val, 0);`
- `console.log(sum); // 10`

57. every()

- Tests if all elements in the array pass a provided function's test.
- **Example:**
- `let arr = [2, 4, 6];`
- `console.log(arr.every(num => num % 2 === 0)); // true`

58.some()

- Tests if at least one element in the array passes a provided function's test.
- **Example:**
- `let arr = [1, 2, 3];`
- `console.log(arr.some(num => num % 2 === 0)); // true`

59.find()

- Returns the first element in the array that satisfies the provided function's condition.
- **Example:**
- `let arr = [1, 2, 3];`
- `let result = arr.find(num => num > 1);`
- `console.log(result); // 2`

60.findIndex()

- Returns the index of the first element in the array that satisfies the provided function's condition.
- **Example:**
- `let arr = [1, 2, 3];`
- `let index = arr.findIndex(num => num > 1);`
- `console.log(index); // 1`